

Syed Hasan Amin Mahmood

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RESEARCH INTERESTS	My research focuses on making AI practicable: learning from limited data, using computation well, and adapting to the people using it. I combine statistical learning and optimization with language modeling and human-AI interaction to understand where resources matter most.	
EDUCATION	Purdue University	
	Ph.D. in Computer Science · GPA: 4.0/4.0	Aug. 2021–Aug. 2026
	Dissertation: <i>From Uniform Optimization to Selective Allocation: Making AI More Practicable across Learning, Generation, and Collaboration.</i>	
	Co-advisors: Ming Yin and Rajiv Khanna .	
	M.S. in Computer Science · GPA: 4.0/4.0	Aug. 2021–May 2024
	Lahore University of Management Sciences (LUMS)	
	B.S. in Electrical Engineering · GPA: 4.0/4.0	Sept. 2016–June 2020
	Minor in Computer Science.	
EXPERIENCE	Doctoral research, Purdue University	2021–2026
	– Learning and optimization. Characterized limits of gradient-based data selection and attribution in deep networks; derived a prompt–response budget rule and an exact law linking learning-rate schedules, weight decay, and scale-invariant training dynamics. [C4] [P1] [P2] – Efficient language generation. Developed consistency training for few-step diffusion generation and examined how sampler calibration changes the quality–efficiency tradeoff. [C8] – Human–AI collaboration. Developed confidence-aware models and adaptive ensembles; analyzed LLM estimates of human perspectives and the effects of AI-assisted critical thinking. [C5] [C2] [C7] [C6]	
	Microsoft, Turing Team	May–Aug. 2025
	<i>Data Scientist (Ph.D.) Intern</i> · Redmond, WA. Developed consistency training for diffusion language generation (ICML 2026). [C8]	
	Microsoft, Turing Team	May–Aug. 2024
	<i>Data Scientist (Ph.D.) Intern</i> · Redmond, WA. Studied LLM alignment through data generation and selection for RLHF and RLAIIF.	
	Afiniti	Oct. 2020–Aug. 2021
	<i>Data Scientist</i> · Lahore, Pakistan. Applied statistical modeling and machine learning to customer–agent pairing.	
	University of Notre Dame	June–Oct. 2020
	<i>Research Assistant</i> · Notre Dame, IN. Studied deep generative models for time-series forecasting in security applications.	

Refereed conference publications

- C8 **Hasan Amin**, Yuan Gao, Yaser Souri, Subhojit Som, Ming Yin, Rajiv Khanna, and Xia Song. **Consistent Diffusion Language Models**. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. Seoul, South Korea, July 2026. [PDF]
- C7 **Hasan Amin**, Harry Yizhou Tian, Xiaoni Duan, Chien-Ju Ho, Rajiv Khanna, and Ming Yin. **From Fallback to Frontline: When Can LLMs be Superior Annotators of Human Perspectives?** *Findings of the Association for Computational Linguistics: ACL 2026*. San Diego, USA, July 2026. [PDF] [Code]
- C6 Harry Yizhou Tian, **Hasan Amin**, and Ming Yin. **Understanding the Effects of AI-Assisted Critical Thinking on Human-AI Decision Making**. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI)*. Barcelona, Spain, April 2026. [PDF]
Best Paper Honorable Mention
- C5 **Hasan Amin**, Ming Yin, and Rajiv Khanna. **Align When They Want, Complement When They Need! Human-Centered Ensembles for Adaptive Human-AI Collaboration**. *Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI)*. Singapore, January 2026. [PDF] [Code]
- C4 **Syed Hasan Amin Mahmood**, Ming Yin, and Rajiv Khanna. **On the Support Vector Effect in DNNs: Rethinking Data Selection and Attribution**. *Proceedings of the 31st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. Toronto, Canada, August 2025. [PDF] [Code]
- C3 Zhuoran Lu, **Syed Hasan Amin Mahmood**, Zhuoyan Li, and Ming Yin. **Mix and Match: Characterizing Heterogeneous Human Behavior in AI-assisted Decision Making**. *Proceedings of the 12th AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*. Pittsburgh, USA, October 2024. [PDF]
- C2 **Syed Hasan Amin Mahmood**, Zhuoran Lu, and Ming Yin. **Designing Behavior-Aware AI to Improve the Human-AI Team Performance in AI-Assisted Decision Making**. *Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI)*. Jeju, South Korea, August 2024. [PDF] [Code]
- C1 **Syed Hasan Amin Mahmood**, Syed Mustafa Ali Abbasi, Ahmed Abbasi, and Fareed Zaffar. **Phishcasting: Deep Learning for Time Series Forecasting of Phishing Attacks**. *Proceedings of the 2020 IEEE International Conference on Intelligence and Security Informatics (ISI)*. Virtual Conference, November 2020. [PDF]

Workshop papers, posters, and extended abstracts

- W3 **Syed Hasan Amin Mahmood** and Rajiv Khanna. **On the Support Vector Effect in DNNs: Rethinking Last Layer Sensitivity-based Instance Attribution**. *NeurIPS Workshop on Attributing Model Behavior at Scale*. New Orleans, LA, December 2023. Preliminary version of [C4].
- W2 **Hasan Amin**, Zhuoran Lu, and Ming Yin. **Give Weight to Human Reactions: Optimizing Complementary AI in Practical Human-AI Teams**. *ICML Workshop on Artificial Intelligence & Human Computer Interaction*. Honolulu, HI, July 2023. [PDF] Preliminary version of [C2].

- W1 **Syed Hasan Amin Mahmood** and Ahmed Abbasi. **Using Deep Generative Models to Boost Forecasting: A Phishing Prediction Case Study**. *Proceedings of the 2020 IEEE International Conference on Data Mining Workshops (ICDMW)*. Sorrento, Italy, November 2020. [PDF]

Preprints

- P3 **Hasan Amin**, Ming Yin, and Rajiv Khanna. **Simple Diffusion Language Models Are More Effective Few-Step Generators Than Reported**. *arXiv preprint arXiv:2609.33947*. September 2026. [PDF]
- P2 **Hasan Amin**, Wei-Kai Chang, and Rajiv Khanna. **When Does Scale-Invariant Optimization Become Unstable? An Exact Schedule Law with Weight Decay**. *arXiv preprint arXiv:2609.09116*. September 2026. [PDF] [Code]
- P1 **Hasan Amin**, Kian Ahrabian, Ming Yin, and Rajiv Khanna. **Escaping the Mode Lottery: Multi-Response Training Improves Language Model Generalization**. *arXiv preprint arXiv:2606.00544*. May 2026. [PDF] [Code]

RESEARCH DATASETS AND RESOURCES

MOSAIC-1K. A benchmark of 1,000 items generated with Claude Opus 4.7 across four domains. Each combines four prompt variants with four responses per variant (16,000 pairs), enabling controlled studies of prompt redundancy and response diversity.

Human and LLM perspective-taking data. New direct judgments and perspective-taking annotations extending a 120-comment toxicity corpus to non-binary participants, alongside repeated LLM estimates and reasoning traces for toxicity and conversational safety. Supports comparisons under matched annotation budgets.

WoofNette. A curated ImageNet subset with 13,500 images spanning familiar objects and challenging dog breeds. Accompanying human-study records provide classifications, confidence ratings, and AI-assisted decisions for investigating complementary AI.

Synthetic college-admissions testbed. Configurable data generators with known decision rules and varying group proportions, plus simulated human expertise and reliance behavior. Enables reproducible comparisons of aligned, complementary, and adaptive AI.

HONORS AND RECOGNITION

Best Paper Honorable Mention , ACM CHI.	2026
Future Leaders of AI , ACM AI Leadership Summit.	2026
Gold Reviewer , ICML.	2026
Special Recognition for Outstanding Review , ACM CHI.	2026
Outstanding Reviewer , ACL Rolling Review, February cycle.	2025
Winner , North America Student Challenge, IEEE Computer Society.	2024
Graduation with High Distinction, LUMS.	2020
Winner , Social Innovation Challenge, LUMS Envision.	2018
Selected for the National University of Singapore Summer Enterprise Program.	2017

Research and travel support

NSF Student Travel Award, ACM SIGKDD.	2025
Graduate Travel Award, Purdue College of Science.	2025
Travel Grant, IJCAI-AIJ.	2024

	Summer Research Grant, Purdue University.	2023
ADDITIONAL PRESENTATIONS	ACM AI Leadership Summit, Future Leaders of AI	Aug. 2026
	<i>Leveraging Data Asymmetries for Design & Optimization of Practicable AI Systems.</i> Doctoral consortium research presentation; Atlanta, GA.	
	IEEE BigData, North America Student Challenge	Dec. 2024
	Presented the winning North America Student Challenge entry; Washington, DC.	
ACADEMIC REVIEWING	Conferences	
	International Conference on Artificial Intelligence and Statistics (AISTATS)	2025, 2026, 2027
	International Conference on Learning Representations (ICLR)	2026, 2027
	Conference on Neural Information Processing Systems (NeurIPS)	2024, 2025, 2026
	International Conference on Machine Learning (ICML)	2025, 2026
	AAAI Conference on Artificial Intelligence	2026, 2027
	ACM Conference on Human Factors in Computing Systems (CHI)	2025, 2026
	Annual Meeting of the Association for Computational Linguistics (ACL)	2025
	ACM Conference on Collective Intelligence (CI)	2025
	ACM SIGKDD Conference on Knowledge Discovery and Data Mining	2024
	International Joint Conference on Artificial Intelligence (IJCAI)	2024
	Journals	
	Transactions on Machine Learning Research (TMLR)	
	International Journal of Human-Computer Interaction (IJHCI)	
	IEEE Intelligent Systems	
	Workshops & other tracks	
	Workshop on GenAI for Health at NeurIPS	2024, 2025, 2026
	ACM CHI Late-Breaking Work	2025
	Workshop on Interpretable Machine Learning in Healthcare at ICML (IMLH)	2022, 2023
LEADERSHIP AND MENTORING	Graduate Committee Representative, Purdue CS	2025–2026
	Represented students on curriculum and policy through the Graduate Student Board.	
	Organizer, Purdue CS Graduate Research Symposium	2025
	Co-organized the department’s inaugural graduate research symposium.	
	Senator for CS, Purdue Graduate Student Government	2024–2025
	Represented CS students in the Senate and on the Graduate Student Board.	
	Member, PGSG Grant Review & Allocation Committee	2024–2025
	Reviewed applications for graduate research and professional development funding.	

	Mentor, Purdue CS Mentor–Mentee Program	2024–2026
	Helped new CS graduate students navigate the transition to graduate study.	
	Mentor, Code to PhD	2025
	Advised underrepresented CS PhD applicants on the graduate application process.	
	Judge, Purdue Spring Undergraduate Research Conference	2025–2026
	Evaluated undergraduate research presentations and provided written feedback.	
TEACHING	Instructor of Record, Purdue University	Fall 2025
	CS 573: Data Mining. Online graduate offering.	
	Graduate Teaching Assistant, Purdue University	Fall 2021–Spring 2025
	– CS 573: Data Mining. – CS 50023: Data Engineering I. – CS 50025: Foundations of Decision Making. – CS 490: Data Science Capstone. – CS 242 / STAT 242: Introduction to Data Science.	
	Teaching Assistant, LUMS	Fall 2018–Fall 2020
	– EE 511: Advanced Digital Signal Processing. – EE 242: Circuits II. – EE 100: Engineering Laboratory. – EE 361: Feedback Control Systems. – MATH 232 / ECON 233: Introduction to Game Theory.	
RESEARCH TOOLS AND LANGUAGES	Programming. Python, C/C++, SQL, R, MATLAB, Bash, LaTeX. ML frameworks. PyTorch, TensorFlow, scikit-learn; Hugging Face Transformers, PEFT. Scientific computing. NumPy, SciPy, pandas, Matplotlib, seaborn. Training and infrastructure. DeepSpeed, Ray, Azure, GCP. Human studies. Experiment design and analysis; Amazon Mechanical Turk, Prolific. Languages. English and Urdu (bilingual); Punjabi and Arabic (beginner).	
SELECTED COURSEWORK	Purdue University. Algorithms; Cognitive Psychology ^{UG} ; Computation & Learning on Graphs; Computational Methods in Optimization; Human–AI Interaction; Human–Computer Interaction; Interpretability in Machine Learning; Natural Language Processing; Probabilistic Causal Inference; Social Psychology ^{UG} ; Statistical Machine Learning; Theoretical CS Toolkit. LUMS. Advanced Digital Signal Processing; Advanced Programming ^{UG} ; Applied Probability; Artificial Intelligence ^{UG} ; Data Science ^{UG} ; Deep Learning ^{UG} ; Dynamic Programming & Reinforcement Learning; Embedded Systems; Information Theory & Machine Learning. UG: undergraduate-level; other courses are graduate-level.	